

Text recognition for a digitised collection

Prepared September 26, 2026 at 9:47 a.m.

Purpose Example: text recognition (OCR) for about 12,000 page images, split into 40 jobs of about 30 minutes each. The figures are assumptions for planning, not measurements of an OCR run.

Lowest emissions

14.0 gCO2e

Stockholm

Highest emissions

2.84 kgCO2e

Calgary

x203

The highest is about 203 times the lowest, for the same work.

Compared 3 of 4 options. Options that cannot be estimated are not ranked.

Option	Where it runs	Jobs × minutes	Energy	Emissions	Cost	Basis
Stockholm	AWS Stockholm Stockholm, Sweden	40 × 30 20 h	656.0 Wh	14.0 gCO2e	US\$11.16	measured
Tokyo	AWS Tokyo Tokyo, Japan	40 × 30 20 h	1.440 kWh	617.8 gCO2e	US\$23.34	portal-estimated
Calgary	AgrifoodTEF AI Data Room Calgary, Alberta, Canada	40 × 30 20 h	8.470 kWh	2.84 kgCO2e	no price registered	portal-estimated
Kashiwa (mdx)	mdx (Information Technology Center, The University of Tokyo) Kashiwa II Campus, Chiba, Japan	40 × 30 20 h	— —		no price registered	cannot be established The power draw of this environment has not been measured yet.

What the labels mean

measured

Power was measured by us on the actual machine.

provider-reported

Stated by the provider. There is no way for us to check it.

portal-estimated

Calculated by this portal from hardware specifications. Not measured.

cannot be established

Not enough is known to give a number, so none is given.

Sources

Electricity

- Sweden (Stockholm): 21.3 gCO2e/kWh

(2025) — Electricity Maps, Grid in Review 2025 — flow-traced (consumption-based, imports included). Production-based on the same page: 19.0. Life-cycle (LCA) basis used by: Stockholm
- Japan (national substitute factor): 429 gCO2e/kWh

(R5年度実績 / FY2023) — Japan Ministry of the Environment / METI substitute factor for electricity suppliers, 0.000429 t-CO2/kWh. Direct emissions at generation. The IEA-based carbon intensity of Japanese generation (2023) is about 485 used by: Tokyo
- Canada, Alberta (Calgary): 335 gCO2e/kWh

(2024) — Government of Alberta. Direct emissions at generation. 2005: 907 → 2019: 629 → 2024: 335 used by: Calgary
- Japan (national substitute factor): 429 gCO2e/kWh

(R5年度実績 / FY2023) — Japan Ministry of the Environment / METI substitute factor, 0.000429 t-CO2/kWh. Not the actual factor of any specific site's electricity contract used by: Kashiwa (mdx)

Power draw

32.8 W measured EC2 g4dn.xlarge / Tesla T4 / average of nvidia-smi inside the job container (2026-08-12, 15 s at 1 Hz). Limit 70 W. Measured during a light workload

72 W portal-estimated Placeholder from the NVIDIA L4 TDP of 72 W. Not measured

350 W portal-estimated Placeholder from the NVIDIA L40S TDP of 350 W. Not measured

Facility overhead (PUE)

AWS Stockholm: 1.0 — facility overhead not counted (lower bound)

AWS Tokyo: 1.0 — facility overhead not counted (lower bound)

AgrifoodTEF AI Data Room: 1.21, stated by the provider

mdx (Information Technology Center, The University of Tokyo): 1.0 — facility overhead not counted (lower bound)

Assumptions and limits

Every option is assumed to take the same time. Faster hardware may finish sooner, which would lower its figures.

Only the electricity used while the job runs is counted. The carbon of building the hardware is not.

CPU power is not included: it cannot be measured from inside a virtual machine.

Emissions use the average carbon intensity of the local grid (location-based). Green-energy certificates are not subtracted.

Sources use different methods: some count the full life cycle of the power station, others only what is emitted when power is generated. Compare regions with that in mind.

SHA-256 of the JSON file

070a544f522679fa3f2186fdf7dad137a1b23f806312aa72f1e942ed6069afa5

If the JSON file you keep gives the same value, it is the same file this report describes.